Exponential-consistency fixed points of parity-relation codes: an exactly solvable binding theory

Felix Robles Elvira (ORCID: 0009-0009-2017-4394; independent researcher)

preprint, not peer reviewed, version 2026-06-11.

Abstract

Let S be a finite set of parity characters ("modes") on n spins and K the binary linear code of relations among them. We study the **exponential-consistency fixed point**: fields h_a on the modes determined self-referentially by $\langle \chi_a \rangle_h = e^{-h_a}$ — each mode's thermal expectation equals its own Boltzmann suppression. The relation-free case is the scalar equation $\tanh h = e^{-h}$, whose root is $\eta = -\ln \theta$ with θ the real root of $\theta^3 + \theta^2 + \theta = 1$ (the reciprocal tribonacci constant). Results, each with machine receipts at 40+ digits: (1) **invariance and decomposition** — the fixed point and its evidence defect depend on S only through the code K, and decompose additively over code components; (2) the **single-relation theorem** — for one weight-w relation the fixed point reduces to a scalar equation, solved exactly: weight 3 *binds* (defect $+0.008438104972291$) while every $w \geq 4$ anti-binds, with the asymptotic law defect $\to \theta^w(1-w\kappa)$, $\kappa = \eta(1-\theta^2)/(1-\theta^2+\theta) = 0.343922878815527$, and sign boundary $w = 1/\kappa = 2.9076$ — so the triangle's binding is decided strictly beyond first order; (3) the **exact defect functional** $\text{defect}(w;\lambda) = \ln(1+\lambda t^w) - w[D(h^*) - D(\eta)]$, $D(h) = he^{-h} - \ln\cosh h$, reproducing the full solved table, and its **second order in closed form**: $d_2(w) = \theta^{2w}\left[-\frac{1}{2} + bw + cw^2\right]$ with $c = \eta(1-\theta^2)^3/(\theta^2\Delta^2)$, $\Delta = 1-\theta^2+\theta$, verified to 10^{-26}; (4) a **frustration identity** (Proposition 5.1; the matched value is computed in an external companion supplied as supplementary material, so this is a proposition with a 14-digit receipt, not a self-contained theorem) — the sign-reversed ($\lambda = -1$) point of the scalar theory equals that companion's oriented frustrated triangle, $+0.201674235292$, to 10^{-14}: frustration is amplitude reversal; (5) **interaction constants** — an exact multi-mode solver for arbitrary codes (validated internally by two independent solution routes and by exact component additivity) gives the shared-mode pair $+0.0208449703908$ and the per-link limit of single-share chains $D_\infty = 0.013273$ (converged to 4×10^{-6} by six links); (6) **algebraic status** — $11\kappa = \eta(7 - 2\theta + \theta^2)$ exactly, while $\kappa \notin \mathbb{Q}(\theta)$ is a theorem (Hermite–Lindemann): the theory's constants are transcendental-form in θ with one η factor. The marginality $\varepsilon = 3\kappa - 1 = 0.0317686364466$ — the triangle's first-order distance from the sign boundary — is the constant exported to companion papers; this paper is its complete, self-contained pedigree.

1. The model

Fix n spins and a set S of m distinct nontrivial parity characters $\chi_a(x) = (-1)^{\langle a,x\rangle}$, $a \in \mathbb{F}_2^n \setminus \{0\}$, $x \in \mathbb{F}_2^n$. The **relation code** $K \subseteq \mathbb{F}_2^S$ is the kernel of $\mathbb{F}_2^S \to \mathbb{F}_2^n$ — a binary linear code of length m, dimension $m - \mathrm{rank}(S)$, and minimum distance ≥ 3 (weight-1 words are trivial characters, weight-2 words repeated characters; both excluded by distinctness). For fields $h \in \mathbb{R}^S$ define

$$Z(h) = \sum_{x\in\mathbb{F}_2^n} \exp\Big(\sum_{a\in S} h_a \chi_a(x)\Big), \qquad \frac{Z(h)}{2^n \prod_a \cosh h_a} = W_K(t) = \sum_{c\in K} \prod_{a\in\mathrm{supp}(c)} t_a,$$

with $t_a = \tanh h_a$ — the standard character (high-temperature) expansion [1], exact; W_K is the code's weight enumerator in the sense of [2].

Definition (exponential consistency). A fixed point is a field h^* solving, for every mode,

$$\langle \chi_a \rangle_h \;=\; e^{-h_a}.$$

The motivation comes from a physics research program in which e^{-h_a} is a record weight and the condition equates a mode's evidence with its weight; the mathematics stands alone, and this paper develops it as such.

Lemma 1.1 (well-posedness, scalar case). For the relation-free mode, $F(h) = \tanh h - e^{-h}$ has $F'(h) = \mathrm{sech}^2 h + e^{-h} > 0$ on all of $\mathbb{R}$: strictly increasing, with $F(0) = -1 < 0 < F(\infty) = 1$ — the root η exists and is unique on $\mathbb{R}$. For the single-relation system of Section 3 at **both** $\lambda = +1$ and $\lambda = -1$ (the latter is the case Proposition 5.1 uses), existence follows from the same sign change (at $\lambda = -1$, $(1-t^2)/(1-t^w) \to 2/w$ gives the positive limit $1 - 2/w$ for $w \geq 3$); uniqueness on $(0, 6)$ is established by *fine sign-change scan* — bracketing on a 20001-point grid with monotone refinement at each bracket — at every $w \leq 40$ and both signs of λ (one root each; receipts; the scan is fine but carries no inter-node derivative bound, so "one root" on $(0, 6)$ is a verified-numerics statement, not a certificate), and roots outside $(0, 6)$ are excluded analytically, split by the sign of λ (the inequality an earlier draft printed here — "$2\tanh^{w-1} h/(1 - \tanh^w h) < 0.01$" — was false, its left side *diverging* as h grows; caught in review): for $h \geq 6$, $e^{-h} \leq e^{-6} < 0.0025$, while at $\lambda = +1$ the relation term is positive so the left side exceeds $\tanh 6 > 0.9999$, and at $\lambda = -1$ the relation term $(1-t^2)\, t^{w-1}/(1-t^w)$ $(t = \tanh h)$ is bounded by its $t \uparrow 1$ limit $2/w \leq 2/3$ for $w \geq 3$ (it increases to that limit in t; verified on a 20000-point sweep for $w \leq 40$), so the left side is at least $\tanh 6 - 2/3 - e^{-6} > 0.33$ from a root — worst margin, at $w = 3$. $h \leq 0$ fails the sign at once. For general multi-mode systems we *assume* the Newton solution continued from $h_a = \eta$ is the relevant fixed point and verify it is locally unique; ruling out asymmetric solutions of symmetric systems globally is open, and the flag is discharged where used (Section 3's reduction): a 300-start random multi-start search per case ($w = 3, 4$; $\lambda = \pm 1$; starts in $(0.05, 4)^w$) finds only the symmetric point, with zero asymmetric solutions (receipts).

The evidence defect. Define, at the fixed point,

$$\mathrm{defect}(S) \;=\; \ln W_K(t^*) \;-\; \sum_a \left[\, D(h_a^*) - D(\eta)\,\right], \qquad D(h) = he^{-h} - \ln\cosh h$$

— equivalently $\mathrm{defect}(S) = D^{\mathrm{free}} - D(P_{h^*}\|U)$ with $D^{\mathrm{free}} = m\, d_0$ (d_0 the per-mode divergence below), the evidence *absorbed* by the relations: a positive defect means the constrained system sits **closer to uniform** than free modes — the relations soak up commitment — and we call such relation sets *binding*; negative = anti-binding. (An earlier draft defined the defect with the opposite sign while printing these values, an inconsistency caught in review; the present definition is the one all printed numbers satisfy, and the gloss above is the correct reading of its sign.)

The free mode. With $K = \{0\}$, each mode separately solves $\tanh h = e^{-h}$. Substituting $x = e^{-h}$ gives $x^3 + x^2 + x = 1$: the root is $\theta = 0.543689012692\ldots$ (the reciprocal tribonacci constant, algebraic of degree 3), with $h = \eta = -\ln\theta = 0.609377863\ldots$ and per-mode divergence $d_0 = \eta\theta - \ln\cosh\eta$.

2. Invariance and decomposition

Theorem 2.1 (code invariance). h^*, Z, and $\mathrm{defect}(S)$ depend on S only through K as a subset structure: mode sets with isomorphic relation codes have equal defects, at any n.

Proof. W_K is a function of K alone; $\psi = \sum_a \ln\cosh h_a + \ln W_K$ generates all expectations; the fixed point and $D(P\|U)$ are functions of ψ. □

Theorem 2.2 (component decomposition). If $K = K_1 \oplus K_2$ over a partition of S (no codeword mixes the parts), then $W = W_1 W_2$, the fixed point decouples, and defect = $\mathrm{defect}_1 + \mathrm{defect}_2$. Modes touched by no codeword contribute exactly zero. □

Receipts: additivity exact across an exhaustive $n = 4$ census and $n = 5$ sampling (861 random relation-carrying sets).

3. The single-relation theorem

For a single codeword of weight w (all w touched modes equal by symmetry, field h), the fixed point is the scalar equation

$$\tanh h \;+\; (1 - \tanh^2 h)\,\frac{\tanh^{w-1} h}{1 + \tanh^w h} \;=\; e^{-h}.$$

Theorem 3.1. Solved at 40 digits: weight 3 **binds**, and w **anti-binds** for every $4 \le w \le 40$ (each case exact); |defect| is maximal at $w = 5$. The full claim — anti-binding for *every* $w \ge 4$ — follows with Corollary 3.2's tail bound:

w	defect(w)	first-order theta^w(1 - w kappa)
3	+0.008438104972291	-0.005105640645752
4	-0.016839123523194	-0.032827182757183
5	-0.023567123903722	-0.034186341316418
6	-0.021780633113267	-0.027469835199371
8	-0.012214241830450	-0.013371668136528
12	-0.002061783361279	-0.002086147023869
16	-0.000262122892964	-0.000262475207817
24	-0.000003228444384	-0.000003228492885
30	-0.000000107108657	-0.000000107108709

Corollary 3.2 (the tail, bounded). For $w > 40$, $\mathrm{defect}(w) = d_1(w)(1 + r_w)$ with $d_1(w) = \theta^w(1 - w\kappa) < 0$ and

$$|r_w| \;\le\; \frac{2c\,w^2\theta^w}{w\kappa - 1},$$

where c is Theorem 4.2's second-order coefficient: by Lemma 1.1's strict monotonicity the fixed-point shift obeys $|h^* - \eta| \le 2(1-\theta^2)\theta^{w-1}/\Delta$ for large w, and the defect's λ-Taylor remainder past first order is then bounded by twice the second-order term, which is $\theta^{2w}\,O(w^2)$ with the explicit coefficient. Grading: the factor-two domination is *not* proved analytically for all w; it is verified against the exact solver — $|r_w| \le$ the displayed bound, with $|r_w|/|d_2/d_1| \le 1$, at every checked $w \in \{6, 12, 40, 60, 80, 100\}$ — so this corollary is a verified-numerics bound, not a theorem. The bound is decreasing in w (as $w^2\theta^w$ is, for $w > 2/\eta \approx 3.3$) and equals 3.0×10^{-9} at $w = 40$, where the true remainder is $|r_{40}| = 1.43 \times 10^{-9}$. (An earlier draft quoted "1.5×10^{-9}" and "0.26" for the bound — those are the second-order *ratios* $|d_2/d_1|$, a mislabeling caught in review; the displayed bound at $w = 6$ is in fact 0.81 and the true remainder there 0.21 — not uniformly small for $6 \le w \le 40$, which is why those cases are solved exactly rather than bounded; the earlier draft's "$\ll$ for $w \ge 6$" claim was wrong and is replaced by this split.) The sign of the tail is that of d_1, i.e. anti-binding. □

Theorem 3.3 (first order and the marginality). The defect's first order in the codeword amplitude is exactly $\theta^w(1 - w\kappa)$ with

$$\kappa = \frac{\eta\,(1-\theta^2)}{1-\theta^2+\theta} = \frac{\eta\,(1+\theta^2)}{2+\theta^2} = 0.343922878815527\ldots$$

The sign boundary sits at $w = 1/\kappa = 2.9076\ldots$: weight 3 lies just above it, at first-order distance

$$\varepsilon \;=\; 3\kappa - 1 \;=\; 0.031768636446582\ldots$$

— the **marginality**. Since the first-order $w = 3$ coefficient is *negative* ($-\varepsilon\theta^3$) while the exact defect is *positive*, the triangle's binding is decided strictly beyond first order: a sign settled by self-interaction cannot be captured by any first-order (weight-counting) law.

4. The exact defect functional and its second order

Theorem 4.1 (the scalar functional). With $h^*(w, \lambda)$ the fixed point of the λ-deformed equation (codeword amplitude λ), the Section 1 definition reduces to

$$\mathrm{defect}(w;\lambda) = \ln(1 + \lambda\tanh^w h^*) - w\,[\,D(h^*) - D(\eta)\,],$$

reproducing every value of Theorem 3.1's table at print precision (worst deviation 4.9×10^{-16}), and cross-checked by solving the fixed point directly on explicit spin realizations (no functional) at $w = 3, 4, 5$ and the Section 6 codes.

Theorem 4.2 (closed-form second order). Expanding $h^* = \eta + \lambda h_1 + \lambda^2 h_2$: $h_1 = -(1-\theta^2)\theta^{w-1}/\Delta$ with $\Delta = 1 - \theta^2 + \theta$; $h_2 = [(1-\theta^2)\theta^{2w-1} - \frac{1}{2}G''h_1^2 - h_1P']/\Delta$ with $G'' = -\theta(3 - 2\theta^2)$ and $P' = (1-\theta^2)\theta^{w-2}[(w-1)(1-\theta^2) - 2\theta^2]$; and with $D'(\eta) = -\eta\theta$, $D''(\eta) = -(2-\eta)\theta - (1-\theta^2)$:

$$d_2(w) = w\theta^{w-1}(1-\theta^2)h_1 - \tfrac{1}{2}\theta^{2w} + w\eta\theta h_2 - \tfrac{w}{2}D''h_1^2 \;=\; \theta^{2w}\Big[-\tfrac{1}{2} + b\,w + c\,w^2\Big],$$

with intercept $-\frac{1}{2}$ exact (the logarithm's own second order),

$$c = \frac{\eta\,(1-\theta^2)^3}{\theta^2\Delta^2} = 0.4625461677\ldots, \qquad b = -0.8775524218\ldots$$

(b given in assembled closed form by the display). *Receipts*: matches high-precision numerical extraction at every $w \in \{3,\ldots,12\}$ to 1.9×10^{-26}; the first-order law re-verified to 10^{-38}.

Warning, kept on record. The expansion is built at $\lambda = 0$; evaluating its truncation at $\lambda = -1$ suggested a value near ε for the sign-reversed triangle — wrong by a factor of six against the exact solution (Section 5). Perturbative truncations of this theory must not be extrapolated to unit amplitude.

5. The frustration identity

Proposition 5.1 (numerical identity; the oriented value's source named). The $\lambda = -1$ (sign-reversed amplitude) point of the scalar theory at $w = 3$ has exact defect

$$\text{defect}(3; -1) = +0.2016742352919823\ldots,$$

equal to the **oriented frustrated triangle** computed independently in the physics program's chirality analysis (corpus Paper 8, "The Tractable Frontier Campaign," supplied as supplementary material with this submission; its printed value +0.201674235292 and frustrated fixed point $h = 1.114142135$ both match — our $\lambda = -1$ root is $h = 1.1141421355$ — and the oriented computation is not reproduced here, which is why this is a proposition with a 14-digit receipt rather than a self-contained theorem). Frustration *is* amplitude reversal: the orientation character's only effect on a single relation is the sign of its codeword term. (The frustrated triangle binds $\sim 24\times$ harder than the unfrustrated one.)

6. Interaction constants from the exact multi-mode solver

For arbitrary codes the fixed point is the coupled system $\langle\chi_a\rangle = e^{-h_a}$, one equation per mode, with defect $= \ln W_K(t^*) - \sum_a [D(h_a^*) - D(\eta)]$. *Validation, internal*: (i) the shared-mode triangle pair is solved twice in this paper — by the general solver and by its symmetric two-class reduction — agreeing at +0.020844970391; (ii) disjoint additivity: the solver's value for the disjoint $w{=}3 \oplus w{=}4$ pair equals defect(3) + defect(4) = 0.008438104972 − 0.016839123523 = −0.008401018551, as Theorem 2.2 requires.

Theorem 6.1 (chain constants). For chains T_k of k triangles, consecutive pairs sharing one mode, the per-link defect increments converge rapidly (successive increment-difference ratios $0.182, 0.176, 0.168$ — mildly drifting, not geometric):

```
D2 = 0.0124068654   D4 = 0.0132458948   D6 = 0.0132724704
D3 = 0.0131167761   D5 = 0.0132686431
per-link limit D_inf = 0.013273 (D6 value; converged to 4e-6,
so the sixth decimal is not significant)
```

Stars (all triangles sharing one common mode) give per-triangle defects $0.0104, 0.0125, 0.0147, 0.0169$ at $k = 2, 3, 4, 5$ — growing, with no per-triangle limit claimed. One caution: the "double-share pair" (two triangles sharing two modes) contains a weight-2 codeword and is therefore *not realizable* by distinct characters in the Section 1 model — it requires the *multiset extension* (allowing repeated characters, i.e. dropping the min-distance-3 exclusion); we report it as abstract-enumerator algebra, flagged. Its raw enumerator defect is $+0.0903175428$; the per-link figure $+0.0818794378$ quoted alongside the chain increments is that value *minus* defect(3) — the increment convention of this section — stated explicitly since a reader reproducing the raw number would otherwise see a discrepancy. These are, to our knowledge, the first interaction constants of this fixed-point theory.

7. Algebraic status of the constants

PSLQ [3] at 40+ digits with symbolic verification (residuals $\sim 10^{-50}$):

$$11\kappa = \eta\,(7 - 2\theta + \theta^2) \quad \text{(exactly; via } (7 - 2\theta + \theta^2)(2 + \theta^2) = 11(1 + \theta^2) \text{ mod the cubic),}$$

and correspondingly $11\varepsilon = -11 + \eta(21 - 6\theta + 3\theta^2)$. Moreover $\kappa \notin \mathbb{Q}(\theta)$ is a *theorem*, not a search result: $\kappa/\eta = (1 + \theta^2)/(2 + \theta^2) \in \mathbb{Q}(\theta) \setminus \{0\}$ exactly, and $\eta = -\ln\theta$ is transcendental by Hermite–Lindemann (θ algebraic, $\neq 0, 1$); if κ were even *algebraic over* $\mathbb{Q}(\theta)$ it would be algebraic over $\mathbb{Q}$ (towers of algebraic extensions are algebraic), forcing $\eta = \kappa \cdot (2 + \theta^2)/(1 + \theta^2)$ algebraic — contradiction. So no algebraic relation over $\mathbb{Q}(\theta)$ holds at any height. The theory's constants are transcendental-form — one η factor over the cubic field.

8. Related work and novelty position

The character expansion is classical (high-temperature expansions; van der Waerden); Ising models on hypergraphs and codes are widely studied, as are self-consistent field equations. What we have not found in the literature: the exponential-consistency condition $\langle \chi_a \rangle = e^{-h_a}$ itself; the exactly solved single-relation family with its sign boundary $1/\kappa$ and the closed-form constants of Sections 3–4; the frustration identity in this form; the chain constants. We state this as a checked absence and commit in advance: if a referee locates any of these, the corresponding section becomes exposition with attribution.

Reproducibility

Every number regenerates from fixed-seed scripts at 40–50 digits (mpmath): the table of Theorem 3.1; the functional and second-order verifications; the frustration identity; the multi-mode solver validations and chain constants; the PSLQ relations with symbolic checks. Bit-identical reruns; code and canonical outputs accompany the submission.

References

[1] B. L. van der Waerden, Die lange Reichweite der regelmässigen Atomanordnung in Mischkristallen, *Z. Phys.* **118** (1941) 473 — character/high-temperature expansions.

[2] F. J. MacWilliams, N. J. A. Sloane, *The Theory of Error-Correcting Codes*, North-Holland (1977) — codes, characters, weight enumerators.

[3] H. R. P. Ferguson, D. H. Bailey, S. Arno, Analysis of PSLQ, *Math. Comp.* **68** (1999) 351 — integer-relation detection.

Companion papers: the physics program exporting $\varepsilon = 3\kappa - 1$ (same author; this paper is the constant's self-contained pedigree).